

Spring 2026 ER Learning Objectives

Lecture HCB

SOM.MK.I.BPM2.2.ER.1.HCB.1001
SOM.MK.I.BPM2.2.ER.1.HCB.1002
SOM.MK.I.BPM2.2.ER.1.HCB.1003
SOM.MK.I.BPM2.2.ER.1.HCB.1004
SOM.MK.I.BPM2.2.ER.1.HCB.1005
SOM.MK.I.BPM2.2.ER.1.HCB.1006
SOM.MK.I.BPM2.2.ER.1.HCB.1007
SOM.MK.I.BPM2.2.ER.1.HCB.1008
SOM.MK.I.BPM2.2.ER.1.HCB.1009
SOM.MK.I.BPM2.2.ER.1.HCB.1010
SOM.MK.I.BPM2.2.ER.1.HCB.1011
SOM.MK.I.BPM2.2.ER.1.HCB.1012
SOM.MK.I.BPM2.2.ER.1.HCB.1013
SOM.MK.I.BPM2.2.ER.1.HCB.1014
SOM.MK.I.BPM2.2.ER.1.HCB.1015
SOM.MK.I.BPM2.2.ER.1.HCB.1016
SOM.MK.I.BPM2.2.ER.1.HCB.1017
SOM.MK.I.BPM2.2.ER.1.HCB.1018
SOM.MK.I.BPM2.2.ER.1.HCB.1019
SOM.MK.I.BPM2.2.ER.1.HCB.1020
SOM.MK.I.BPM2.2.ER.1.HCB.1021
SOM.MK.I.BPM2.2.ER.1.HCB.1022
SOM.MK.I.BPM2.2.ER.1.HCB.1023
SOM.MK.I.BPM2.2.ER.1.HCB.1024
SOM.MK.I.BPM2.2.ER.1.HCB.1025
SOM.MK.I.BPM2.2.ER.1.HCB.1026
SOM.MK.I.BPM2.2.ER.1.HCB.1027
SOM.MK.I.BPM2.2.ER.1.HCB.1028
SOM.MK.I.BPM2.2.ER.1.HCB.1029
SOM.MK.I.BPM2.2.ER.1.HCB.1030
SOM.MK.I.BPM2.2.ER.1.HCB.1031

Flipped Class: Integrative Histology of the Endocrine System

Classify endocrine glands based on chemical components of their secretion.
Describe the basic histological characteristics of all endocrine glands.
Describe the development of the pituitary gland in terms of its cellular and neuronal origins.
Identify histologically the glandular and nervous tissue components of the pituitary gland
Discuss the role of the hypothalamus in the regulation of the pituitary gland activity
List the components of the hypothalamohypophyseal tract.
Discuss the significance of the hypothalamohypophyseal tract in terms of hormone production and secretion.
List the vascular components of the hypothalamohypophyseal portal system.
Discuss the significance of the hypothalamohypophyseal portal system in terms of hormone production and secretion.
List the hormones secreted by the anterior pituitary
Describe the microscopic structures of anterior pituitary.
Briefly discuss the main effects of the hormones produced by the anterior pituitary on their target tissues.
Describe the location of pars intermedia.
Describe the microscopic structure of pars intermedia.
List the two (2) hormones that are stored in the posterior pituitary.
Describe the microscopic structure of the posterior pituitary.
Give the origin of the two (2) hormones that are stored in the posterior pituitary.
Discuss the main effect(s) of the two (2) hormones that are stored in the posterior pituitary on their target tissue.
Discuss pituitary adenomas and possible clinical presentations.
Describe the anatomical location of the adrenal gland.
Describe the blood supply to the adrenal gland.
List the cortical zones of the adrenal gland.
Describe the microscopic structure (including the cell types) of the cortical zones of the adrenal gland.
Give the hormones produced in each of the cortical zones of the adrenal gland.
Briefly discuss the main effect(s) of the hormones in each of the cortical zones of the adrenal gland on their target tissue(s).
Identify the medulla of the adrenal gland.
Describe the microscopic structure of the medulla of the adrenal gland.
Give the hormone(s) produced in the medulla of the adrenal gland.
Briefly discuss the main effect of the hormones produced in the medulla of the adrenal gland on their target tissues.
Compare and contrast Cushing's disease vs. Cushing's syndrome.
Describe the anatomical location of the pancreas.

SOM.MK.I.BPM2.2.ER.1.HCB.1032	Describe the blood supply to pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1033	Briefly distinguish between the exocrine and the endocrine pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1034	Identify the pancreatic islets of the endocrine pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1035	List the cells of the pancreatic islets of the endocrine pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1036	Discuss the microscopic structure of the pancreatic islets of the endocrine pancreas, including the location of each cell type found within the islets.
SOM.MK.I.BPM2.2.ER.1.HCB.1037	Briefly discuss the main effects of the hormones secreted by the cells of the pancreatic islets on their target tissues.
SOM.MK.I.BPM2.2.ER.1.HCB.1038	Describe the anatomical location of the thyroid gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1039	State the function(s) of the thyroid gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1040	Describe the microscopic appearance of the thyroid gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1041	Describe the microscopic changes seen in the thyroid gland in hypothyroidism.
SOM.MK.I.BPM2.2.ER.1.HCB.1042	Describe the changes seen in the thyroid gland in hyperthyroidism.
SOM.MK.I.BPM2.2.ER.1.HCB.1043	Give the localization of parafollicular cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1044	Discuss the function of parafollicular cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1045	Describe the microscopic appearance of the parafollicular cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1046	Briefly discuss the main effect of the hormone secreted by the parafollicular cells on their target tissue(s).
SOM.MK.I.BPM2.2.ER.1.HCB.1047	Give the anatomical location of the parathyroid glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1048	Discuss the function of the parathyroid glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1049	Describe the microscopic appearance (including the 2 cell types) of the parathyroid glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1050	Briefly discuss the main effect of the hormone secreted by the main cell type of the parathyroid glands on their target tissue(s).
SOM.MK.I.BPM2.2.ER.1.HCB.1051	Analyze the relationship between the parathyroid glands and the parafollicular cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1052	Give the localization of the pineal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1053	Give the function(s) of the pineal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1054	Describe the microscopic structure (including the 2 cell populations) of the pineal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1055	Discuss the clinical significance of corpora arenacea.
SOM.MK.I.BPM2.2.ER.1.HCB.1056	Briefly discuss the main effect of the major hormone secreted by the pineal gland on its target tissue(s).

Lecture PHYS	Hypothalamus and Anterior Pituitary Functions
SOM.MK.II.BPM2.1.ER.1.PHYS.1003	Explain the relationship between anterior pituitary and the hypothalamus and the regulation of hypothalamic hormone release.
SOM.MK.II.BPM2.1.ER.1.PHYS.1004	Identify appropriate hypothalamic factors that control the secretion of each of the anterior pituitary hormones.
SOM.MK.II.BPM2.1.ER.1.PHYS.1005	List the different cell types that synthesize and secrete anterior pituitary hormones.
SOM.MK.II.BPM2.1.ER.1.PHYS.1140	Explain the chemical nature and clinical significance of the three different types of anterior pituitary hormones and their functional relationship with placental hormones.
SOM.MK.II.BPM2.1.ER.1.PHYS.1006	Understand negative feedback control of anterior pituitary hormone secretion at multiple levels.

SOM.MK.II.BPM2.1.ER.1.PHYS.1007	Describe growth hormone, which 2nd messenger it activates and what triggers its release.
SOM.MK.II.BPM2.1.ER.1.PHYS.1008	Describe the effects of GHRH and somatostatin on GH release and their mechanism of action on target using which 2nd messenger.
SOM.MK.II.BPM2.1.ER.1.PHYS.1009	Understand the pattern of secretion growth hormone
SOM.MK.II.BPM2.1.ER.1.PHYS.1010	Describe the metabolic and growth promoting actions of growth hormone. Describe their mechanism of action on target organs
SOM.MK.II.BPM2.1.ER.1.PHYS.1011	Describe the relationship between growth hormone and the insulin-like growth factors and their binding proteins in the regulation of growth. Explain nutritional and other factors that affect growth hormone and somatic growth. Predict the developmental effects of too much or too little growth hormone.
SOM.MK.II.BPM2.1.ER.1.PHYS.1012	
SOM.MK.II.BPM2.1.ER.1.PHYS.1013	Discuss the feedback mechanisms in the regulation of growth hormone.
SOM.MK.II.BPM2.1.ER.1.PHYS.1141	Explain additional factors that stimulate or inhibit growth hormone secretion.

Lecture PHYS	Anterior and Posterior Pituitary Functions
SOM.MK.II.BPM2.1.ER.1.PHYS.1014	Describe the effects of loss of all pituitary hormones and effects of injury to pituitary stalk.
SOM.MK.II.BPM2.1.ER.1.PHYS.1015	Describe the chemical nature, synthesis and release of prolactin
SOM.MK.II.BPM2.1.ER.1.PHYS.1016	Explain the mechanism of action of prolactin at its target cells and list its physiological functions on 2 key target tissues.
SOM.MK.II.BPM2.1.ER.1.PHYS.1017	Describe symptoms of prolactinomas and prolactin imbalance
SOM.MK.II.BPM2.1.ER.1.PHYS.1018	List 2 posterior pituitary hormones and discuss their structural similarity.
SOM.MK.II.BPM2.1.ER.1.PHYS.1019	Explain the regulation of vasopressin secretion.
SOM.MK.II.BPM2.1.ER.1.PHYS.1020	List the 2 target cells for vasopressin and explain why vasopressin is also known as antidiuretic hormone.
SOM.MK.II.BPM2.1.ER.1.PHYS.1021	Identify disease states caused by a) over secretion, and b) under secretion of vasopressin and list the principal symptoms of each.
SOM.MK.II.BPM2.1.ER.1.PHYS.1022	Name the stimuli for oxytocin release in relation to its reproductive and lactation functions.
SOM.MK.II.BPM2.1.ER.1.PHYS.1023	List the 2 target organs and the functional effects of oxytocin.

DLA PHYS	Nutrition & Aging
SOM.MK.III.BPM2.2.ER.3.BCHM.1001	Describe changes in physiological, biochemical and endocrine functions associated with aging
SOM.MK.III.BPM2.2.ER.3.BCHM.1002	Identify the nutritional changes related to aging. Review nutrient requirements for the elderly including energy, protein, fat and cholesterol, carbohydrates, dietary fiber, water, vitamin B12, folate, calcium and vitamin D
SOM.MK.III.BPM2.2.ER.3.BCHM.1003	
SOM.MK.III.BPM2.2.ER.3.BCHM.1004	Describe the risk factors for poor nutrition in the elderly including poor appetite, immobility, concurrent illness, and decreased nutrient absorption Explain how under-nutrition exacerbates disease and leads to poor recovery from illness and hospitalization and describe how disease and hospitalization can lead to under-nutrition
SOM.MK.III.BPM2.2.ER.3.BCHM.1005	
SOM.MK.III.BPM2.2.ER.3.BCHM.1006	Review how the aging process can be accelerated by oxidative and inflammatory stress

SOM.MK.III.BPM2.2.ER.3.BCHM.1007

SOM.MK.III.BPM2.2.ER.3.BCHM.1008

Define frailty and sarcopenia. Review metabolic changes associated with aging highlighting negative nitrogen balance and loss of muscle protein

Review protein energy malnutrition in the elderly. Describe methods for identification and management

Lecture BCHM

Synthesis of Amino Acid Derivative Hormones

SOM.MK.II.BPM2.1.ER.2.BCHM.1010

SOM.MK.II.BPM2.1.ER.2.BCHM.1011

SOM.MK.II.BPM2.1.ER.2.BCHM.1012

SOM.MK.II.BPM2.1.ER.2.BCHM.1013

Discuss the biosynthesis and degradation of hormones that are derived from amino acids and analyze clinical lab tests that rely on the relevant molecule.

Review the pathways for synthesis of the hormones derived from amino acids

Describe the physiological role of hormones derived from amino acid precursors.

Identify hormones formed from tyrosine and explain the biosynthesis of catecholamines.

Highlight the role of enzymes, coenzymes [tetrahydrobiopterin (BH4), PLP, vitamin C (ascorbic acid)] and S-adenosyl methionine (SAM) in the synthesis of

SOM.MK.II.BPM2.1.ER.2.BCHM.1014

SOM.MK.II.BPM2.1.ER.2.BCHM.1015

Dopamine, Norepinephrine and Epinephrine.

Explain the role of MAO and COMT in degradation of catecholamines

Describe VMA, metanephrine and homovanillic acid as the degradation products of the catecholamines and their clinical significance in the diagnosis of pheochromocytoma

SOM.MK.II.BPM2.1.ER.2.BCHM.1016

SOM.MK.II.BPM2.1.ER.2.BCHM.1017

SOM.MK.II.BPM2.1.ER.2.BCHM.1018

SOM.MK.II.BPM2.1.ER.2.BCHM.1019

SOM.MK.II.BPM2.1.ER.2.BCHM.1020

SOM.MK.II.BPM2.1.ER.2.BCHM.1021

SOM.MK.II.BPM2.1.ER.2.BCHM.1022

Describe the role of thyroglobulin, dietary iodine and post-translational iodination of thyroid hormones to form T3 and T4.

Outline and explain the biosynthesis of T3 and T4 in the thyroid gland.

Identify hormones and neurotransmitters formed from tryptophan.

Indicate the steps in the formation of serotonin (5-hydroxy tryptamine; 5-HT) from tryptophan

Explain the formation of 5-HIAA formation (Role of MAO) and analyze its clinical significance

Indicate the precursor molecule of melatonin and identify its significance in the pineal gland

Correlate the following disorders to the biochemical basis of the manifestations and the associated laboratory findings: Parkinson disease;

SOM.MK.II.BPM2.1.ER.2.BCHM.1023

Pheochromocytoma; Carcinoid syndrome.

Small group HPWP

ER SG 01 - BPM 2 HPWP 01 - Wellness, Healthy Mind & Healthy Body

SOM.PB.1.BPM2.HPWP.026

SOM.PB.1.BPM2.HPWP.027

SOM.PB.1.BPM2.HPWP.028

Apply self-assessment tools, such as the Depression, Anxiety, Stress Scale (DASS) and Sleep Quality Scale (SQS), in the domains of stress, anxiety, depression, physical activity, and sleep to design a holistic, individualized treatment plan

Apply behavioral interventions that include mindfulness, progressive muscle relaxation, and good sleep hygiene designed to decrease stress and increase self-health-promoting behaviors.

Illustrate why behavioral changes, such as increasing physical activity, can be difficult, using a case study about a patient who has to make health-related changes, and self-reflection on what challenges they may face.

Small group HPWP

ER SG 02 - BPM2 HPWP 02 - Professionalism

SOM.PB.5.BPM2.HPWP.029	Explain how professional communication, including the ability to summarize complex medical terminology, enhances effective patient education across health disciplines
SOM.CS.1.BPM2.HPWP.030	Develop professional communication techniques for negotiating collaborative solutions.
SOM.CS.1.BPM2.HPWP.031	Analyze the effectiveness of verbal and non-verbal communication skills through role playing scenarios.
SOM.PB.2.BPM2.HPWP.032	Identify challenges to professionalism, and explore strategies for managing them, recognizing that conflicts often involve multiple perspectives and interpretations.
SOM.PB.5.BPM2.HPWP.033	Discuss the importance of the roles played by all members of the healthcare team, including nurses, midwives and support staff in delivering effective patient care
SOM.PB.1.BPM2.HPWP.034	Reflect on the importance of being open to diverse perspectives in creating a shared plan forward.

Lecture BCHM	Steroid Hormone Metabolism
SOM.MK.II.BPM2.1.ER.2.BCHM.1024	Review the classes of steroid hormones (glucocorticoids, mineralocorticoids, androgens, estrogens and progestins). Indicate the steroid hormones synthesized by the different adrenal cortex layers (zona fasciculata, z. glomerulosa and z. reticularis) and the role of ACTH in their release.
SOM.MK.II.BPM2.1.ER.2.BCHM.1025	Review the synthesis and the metabolic actions of aldosterone. Indicate the role of angiotensin.
SOM.MK.II.BPM2.1.ER.2.BCHM.1026	Review the synthesis and the metabolic actions of cortisol.
SOM.MK.II.BPM2.1.ER.2.BCHM.1027	Indicate the synthesis of DHEA and androstenedione in adrenal cortex. Describe the synthesis of testosterone and estrogen from DHEA and androstenedione in peripheral tissues.
SOM.MK.II.BPM2.1.ER.2.BCHM.1028	Discuss steroid hormone synthesis related to StAR and desmolase (CYP11A). Outline steroid synthesis in the adrenal cortex.
SOM.MK.II.BPM2.1.ER.2.BCHM.1029	Differentiate congenital adrenal hyperplasias (CAH) in patients with deficiencies of 3-hydroxysteroid DH, CYP 21, CYP 11B and CYP 17 based on clinical features and laboratory findings.
SOM.MK.II.BPM2.1.ER.2.BCHM.1030	
SOM.MK.II.BPM2.1.ER.2.BCHM.1031	Describe the synthesis of testosterone in testes and of estradiol in ovaries. Discuss the functions of 5-alpha reductase and aromatase (CYP 19)
SOM.MK.II.BPM2.1.ER.2.BCHM.1032	Indicate the hormone abnormality in: Cushing syndrome and Addison disease.
SOM.MK.II.BPM2.1.ER.2.BCHM.1033	Indicate the biochemical basis for the occurrence of hyperglycemia in Cushing syndrome.

Lecture PHYS	Adrenal Glands
SOM.MK.II.BPM2.1.ER.1.PHYS.1024	Identify the functional zones (one medullary and three cortical zones)of the adrenal glands and the principal hormones secreted from each zone, and significance of distribution of blood supply
SOM.MK.II.BPM2.1.ER.1.PHYS.1025	Understand the differential regulation of cortisol versus aldosterone release, chemical nature, mechanism of actions and release of CRH & ACTH
SOM.MK.II.BPM2.1.ER.1.PHYS.1026	Explain biological actions of cortisol and adrenal androgens. Identify the causes and consequences of over secretion and under secretion of cortisol.
SOM.MK.II.BPM2.1.ER.1.PHYS.1027	Describe the mechanism through which angiotensin II and high K+ stimulate aldosterone synthesis in the cells of the zona glomerulosa.

SOM.MK.II.BPM2.1.ER.1.PHYS.1028	List the one major mineralocorticoids (aldosterone) and identify their biological actions and target organs or tissues, and its role sodium and potassium excretion.
SOM.MK.II.BPM2.1.ER.1.PHYS.1029	Identify the causes and consequences of over secretion and under secretion of mineralocorticoids.

Lecture PHYS

SOM.MK.II.BPM2.1.ER.1.PHYS.1049	Illustrate the structure of the thyroid gland and explain the mechanism of action of TRH and TSH on their target organs.
SOM.MK.II.BPM2.1.ER.1.PHYS.1051	Describe the importance of iodine in diet, its absorption, uptake, distribution, and excretion.
SOM.MK.II.BPM2.1.ER.1.PHYS.1053	Describe the physiologic effects and mechanisms of action of thyroid hormones and clinical applications.
SOM.MK.II.BPM2.1.ER.1.PHYS.1054	Explain how thyroid hormone is transported, metabolized and regulated
SOM.MK.II.BPM2.1.ER.1.PHYS.1055	List and explain the causes of thyroid gland enlargement. Predict the consequences of thyroid hromone under- and over-secretion.

Lecture BCHM

SOM.MK.II.BPM2.1.ER.2.BCHM.1034	Distinguish between the types of diabetes: type 1 and type 2 (differences in insulin secretion) and secondary diabetes mellitus. Explain the role of free fatty acids, leptin, adiponectin, incretins and proinflammatory cytokines in development of insulin resistance in obesity and type 2 diabetes mellitus Correlate presenting features of type 1 and type 2 diabetes mellitus and basis for clinical features Evaluate changes in serum lipids (dyslipidemia) in diabetes mellitus and correlate to occurrence of macrovascular changes.
SOM.MK.II.BPM2.1.ER.2.BCHM.1035	
SOM.MK.II.BPM2.1.ER.2.BCHM.1036	
SOM.MK.II.BPM2.1.ER.2.BCHM.1037	
SOM.MK.II.BPM2.1.ER.2.BCHM.1038	Define criteria for diagnosis of diabetes mellitus: Fasting plasma glucose, 2-hour plasma glucose following oral glucose challenge, HbA1c, role of OGTT Develop the concept of medical nutrition therapy; Highlight the significance of dietary carbohydrates, artificial sweeteners, Glycemic index, Dietary lipids and Mediterranean diet, Dietary fiber in diabetics
SOM.MK.II.BPM2.1.ER.2.BCHM.1039	
SOM.MK.II.BPM2.1.ER.2.BCHM.1040	Evaluate significance of exercise and weight management in management of patients with prediabetes, metabolic syndrome and diabetes mellitus
SOM.MK.II.BPM2.1.ER.2.BCHM.1041	Identify disease states caused by: a) over secretion, b) under secretion of insulin, or c) decreased sensitivity to insulin, D) hypoglycemia
SOM.MK.II.BPM2.1.ER.2.BCHM.1043	Summarize the release of insulin-counterregulatory hormones like glucagon, epinephrine and cortisol.
SOM.MK.II.BPM2.1.ER.2.BCHM.1044	Describe the adrenergic and neuroglycopenic symptoms in patients with hypoglycemia
SOM.MK.II.BPM2.1.ER.2.BCHM.1045	Differentiate hypoglycemia due to a. Injection of insulin. b. Reactive (postprandial) hypoglycemia. c. Alcohol induced e. Insulinoma. f. Factitious (Insulin injection, sulfonylurea).
SOM.MK.II.BPM2.1.ER.2.BCHM.1046	Justify the significance of estimation of serum insulin, C-peptide, prosinsulin in the differential diagnosis of various types of hypoglycemia

Required reading BCHM

SOM.MK.II.BPM2.1.ER.2.BCHM.1047	Identify the major hormones secreted from the endocrine pancreas, their cells of origin, and their chemical nature
SOM.MK.II.BPM2.1.ER.2.BCHM.1048	Describe the major stimulus for synthesis, secretion and regulation of insulin

SOM.MK.II.BPM2.1.ER.2.BCHM.1049
SOM.MK.II.BPM2.1.ER.2.BCHM.1050
SOM.MK.II.BPM2.1.ER.2.BCHM.1052
SOM.MK.II.BPM2.1.ER.2.BCHM.1053

Describe the hormonal changes after a carbohydrate-rich meal and the function of these hormones.
Explain insulin formation and secretion by the pancreatic beta- cells. Highlight the significance of C-peptide.
Describe the regulation of glucagon secretion.
Describe the principal actions of glucagon on target tissues including intracellular signal transduction.

Small group HCB;BCHM;PHYS
SOM.MK.II.BPM2.1.ER.2.BCHM.1415
SOM.MK.II.BPM2.1.ER.2.BCHM.1416

ER SG 03 - Endocrine
Review the pathway of steroid hormone synthesis.
Describe the synthesis of thyroid hormones and the role of iodine in this pathway.

SOM.MK.II.BPM2.1.ER.1.PHYS.1400
SOM.MK.II.BPM2.1.ER.1.PHYS.1401
SOM.MK.II.BPM2.1.ER.1.PHYS.1402

Evaluate the relationship between anterior pituitary and the hypothalamus and explain how it is useful in clinical diagnosis of their dysfunctions.
Apply the biological actions of cortisol and adrenal androgens in explaining the clinical presentations of Cushing syndrome.
Predict the consequences of under and oversecretion of the thyroid hormones.
Compare and contrast clinical features, serum and urinary alterations of calcium, phosphate and vitamin D, and relate them to the alterations in serum parathyroid hormone concentrations.

SOM.MK.II.BPM2.1.ER.1.PHYS.1403
SOM.MK.I.BPM2.2.ER.1.HCB.1057
SOM.MK.I.BPM2.2.ER.1.HCB.1058
SOM.MK.I.BPM2.2.ER.1.HCB.1059
SOM.P.I.BPM2.2.ER.6.HCB.1060

List the cells, tissues, organs and associated structures of the endocrine system.
Present and demonstrate the cells, tissues, organs and associated structures of the endocrine system to the group.
Apply the histology of the endocrine system to clinical cases and correlations.
Develop both professional skills and presentation skills.

Lecture PHYS
SOM.MK.II.BPM2.1.ER.1.PHYS.1056
SOM.MK.II.BPM2.1.ER.1.PHYS.1057
SOM.MK.II.BPM2.1.ER.1.PHYS.1058
SOM.MK.II.BPM2.1.ER.1.PHYS.1059

Calcium Homeostasis, PTH & Vitamin D
Identify the normal range of dietary calcium intake, calcium distribution in the body, and routes of calcium excretion.
Identify the normal range of dietary phosphate intake, phosphate distribution in the body, and routes of phosphate excretion.
Describe the organ that secretes parathyroid hormone, its chemical nature and its mechanism of action
Describe the regulation of parathyroid hormone secretion and the role of the calcium sensing receptor.
Identify the sources of vitamin D and diagram the biosynthetic pathway and the organs involved in modifying it to the biologically active 1,25(OH₂)D₃ (1-25 dihydroxy cholecalciferol).

SOM.MK.II.BPM2.1.ER.1.PHYS.1060
SOM.MK.II.BPM2.1.ER.1.PHYS.1061
SOM.MK.II.BPM2.1.ER.1.PHYS.1062
SOM.MK.II.BPM2.1.ER.1.PHYS.1063
SOM.MK.II.BPM2.1.ER.1.PHYS.1064

Identify the target organs and cellular mechanisms of action for vitamin D.
Describe calcitonin and its role in humans
Explain the clinically relevant consequences of oversecretion undersecretion of parathyroid hormone.
Describe the consequences of vitamin D deficiency and vitamin D excess.

Lecture ANAT
SOM.MKII.BPM2.1.ER.1.ANAT.1102

Clinical Anatomy and Imaging of the Reproductive system - Male Reproductive Organs
Identify the major male pelvic organs, including the bladder, prostate, seminal vesicles, vas deferens, urethra, and rectum.

SOM.MKII.BPM2.1.ER.1.ANAT.1103	Describe the anatomical relationships of the male pelvic organs to each other and surrounding structures, including the pelvic diaphragm and the bony pelvis
SOM.MKII.BPM2.1.ER.1.ANAT.1104	Identify the male pelvic organs on medical imaging studies such as MRI, CT scans, cystourethrography, vesiculography etc.
SOM.MKII.BPM2.1.ER.1.ANAT.1105	Describe the location of the testes within the scrotum and their relationship to the epididymis and spermatic cord.
SOM.MKII.BPM2.1.ER.1.ANAT.1106	Describe the general anatomy of the testis including the testicular coverings (tunica vasculosa, tunica albuginea and tunica vaginalis) and the testicular lobules
SOM.MKII.BPM2.1.ER.1.ANAT.1107	Describe the anatomical relationship of the vas deferens to the seminal vesicles, bladder and ureters
SOM.MKII.BPM2.1.ER.1.ANAT.1108	Trace the pathway of the vas deferens from the epididymis through the inguinal canal, into the pelvic cavity, and to its termination at the ejaculatory ducts.
SOM.MKII.BPM2.1.ER.1.ANAT.1109	Relate the anatomy of the vas deferens to surgical procedures such as vasectomy and its reversal.
SOM.MKII.BPM2.1.ER.1.ANAT.1040	Describe the relationship of the prostate gland with the ampulla of the ductus deferens, seminal vesicles, bladder, urethra, rectum and pelvic floor.
SOM.MKII.BPM2.1.ER.1.ANAT.1110	Describe the zones of the prostate (peripheral, central, transitional, anterior fibromuscular stroma).
SOM.MKII.BPM2.1.ER.1.ANAT.1111	Describe the location of the seminal vesicles posterior to the bladder and their relationship to the rectum, vas deferens, and prostate.
SOM.MKII.BPM2.1.ER.1.ANAT.1112	Describe the role of the seminal vesicles in the production of seminal fluid and their connection to the ejaculatory ducts.
SOM.MKII.BPM2.1.ER.1.ANAT.1113	Describe the anatomical location, structure and function of the bulbourethral glands
SOM.MKII.BPM2.1.ER.1.ANAT.1047	Describe the blood supply, venous and lymphatic drainage of the prostate gland, ductus deferens, seminal vesicles, bulbourethral glands, testes, scrotum, penis
SOM.MKII.BPM2.1.ER.1.ANAT.1134	Describe testicular torsion, its signs, symptoms and clinical significance
SOM.MKII.BPM2.1.ER.1.ANAT.1049	Describe the origin, and type of nerve fibers, found in the superior hypogastric plexus and the inferior hypogastric (pelvic) plexus
SOM.MKII.BPM2.1.ER.1.ANAT.1038	Compare and contrast the visceral pain pathways for structures above the peritoneal floor of the pelvis versus structures below the peritoneal floor (pelvic pain line) in males
SOM.MKII.BPM2.1.ER.1.ANAT.1114	Describe the autonomic innervation of the male pelvic organs and the relationship to the pelvic pain line
SOM.MKII.BPM2.1.ER.1.ANAT.1115	Relate the zonal anatomy of the prostate to conditions such as benign prostatic hyperplasia (BPH) and prostate cancer.
SOM.MKII.BPM2.1.ER.1.ANAT.1036	Explain how fluid in the abdomen can reach the male pelvic peritoneal pouches and where it would ultimately settle.
SOM.MKII.BPM2.1.ER.1.ANAT.1043	Describe the anatomical basis for the symptoms most commonly involved in benign prostatic enlargement and prostate cancer.
SOM.MKII.BPM2.1.ER.1.ANAT.1044	Explain why prostate cancer usually leads to erectile and urinary dysfunction. Identify the nerves involved in this process.
SOM.MKII.BPM2.1.ER.1.ANAT.1045	Describe the surgical procedure called TURP (Trans-Urethral Resection of the Prostate) and common postoperative complications resulting from this procedure.
SOM.MKII.BPM2.1.ER.1.ANAT.1048	List the common sites of prostate cancer metastasis and explain how prostate cancer cells can travel to the central nervous system.
DLA ANAT	Autonomic Innervation of the Male Pelvic Organs
SOM.MKII.BPM2.1.ER.1.ANAT.1049	Describe the origin, and type of nerve fibers, found in the superior hypogastric plexus and the inferior hypogastric (pelvic) plexus
SOM.MKII.BPM2.1.ER.1.ANAT.1114	Describe the autonomic innervation of the male pelvic organs and the relationship to the pelvic pain line

SOM.MKII.BPM2.1.ER.1.ANAT.1038

Lecture ANAT

- SOM.MKII.BPM2.1.ER.1.ANAT.1001
- SOM.MKII.BPM2.1.ER.1.ANAT.1002
- SOM.MKII.BPM2.1.ER.1.ANAT.1003
- SOM.MKII.BPM2.1.ER.1.ANAT.1004
- SOM.MKII.BPM2.1.ER.1.ANAT.1117

- SOM.MKII.BPM2.1.ER.1.ANAT.1012
- SOM.MKII.BPM2.1.ER.1.ANAT.1118
- SOM.MKII.BPM2.1.ER.1.ANAT.1006
- SOM.MKII.BPM2.1.ER.1.ANAT.1005
- SOM.MKII.BPM2.1.ER.1.ANAT.1009
- SOM.MKII.BPM2.1.ER.1.ANAT.1013
- SOM.MKII.BPM2.1.ER.1.ANAT.1015
- SOM.MKII.BPM2.1.ER.1.ANAT.1014
- SOM.MKII.BPM2.1.ER.1.ANAT.1130
- SOM.MKII.BPM2.1.ER.1.ANAT.1017
- SOM.MKII.BPM2.1.ER.1.ANAT.1050
- SOM.MKII.BPM2.1.ER.1.ANAT.1046
- SOM.MKII.BPM2.1.ER.1.ANAT.1019
- SOM.MKII.BPM2.1.ER.1.ANAT.1018
- SOM.MKII.BPM2.1.ER.1.ANAT.1132

Lecture HCB

- SOM.MK.I.BPM2.2.ER.1.HCB.1077
- SOM.MK.I.BPM2.2.ER.1.HCB.1078
- SOM.MK.I.BPM2.2.ER.1.HCB.1079
- SOM.MK.I.BPM2.2.ER.1.HCB.1080
- SOM.MK.I.BPM2.2.ER.1.HCB.1081
- SOM.MK.I.BPM2.2.ER.1.HCB.1082
- SOM.MK.I.BPM2.2.ER.1.HCB.1083

Compare and contrast the visceral pain pathways for structures above the peritoneal floor of the pelvis versus structures below the peritoneal floor (pelvic pain line) in males

Clinical Anatomy and Imaging of the Reproductive system - Male Perineum

- Describe the boundaries and divisions of the perineum.
- List the boundaries of the urogenital triangle.
- Describe the perineal membrane.
- Describe the perineal body, its location and clinical significance in the perineum.
- Describe the contents of the male urogenital triangle
- Describe superficial perineal fascia (Colles’ fascia) in males and its relationship with the superficial fascia of the abdomen and scrotum (Dartos fascia).
- Describe the deep perineal fascia in males
- List the boundaries, contents and innervation of the superficial perineal pouch in males
- List the boundaries, contents and innervation of the deep perineal pouch in males
- Describe the structure and innervation of male external genitalia in including the scrotum, erectile bodies and penis
- Describe the arterial supply, venous drainage and lymphatic drainage of the male perineum.
- Describe the course of the internal pudendal artery and its branches within the male perineum
- Describe the course of the pudendal nerve and its branches within the male perineum.
- Describe the innervation of the male perineum
- Describe the nervous pathways for control of erection including the function of the cavernous nerves
- Describe the nervous pathways for control of emission and ejaculation.
- Describe the location and clinical significance of anterior and posterior urethral tears
- Explain the extravasation of urine in damage to the male urethra, with and without damage to Buck’s fascia.
- Explain why extravasated urine will not pass into the thighs or the ischioanal fossa.
- Identify the parts of the male perineum on imaging studies such as CT scans and MRI

Histology of the Male Reproductive System

- Identify the locations of the testicular connective tissue coverings (tunica vaginalis, tunica albuginea).
- Distinguish between the histological composition of the testicular connective tissue coverings (tunica vaginalis, tunica albuginea).
- State the significance of the testicular connective tissue coverings (tunica vaginalis, tunica albuginea).
- State the significance of testicular lobules.
- Give the function of the mediastinum testis.
- Describe the microscopic structure of the mediastinum testis.
- List the functions of the seminiferous tubules

SOM.MK.I.BPM2.2.ER.1.HCB.1084	Describe the microscopic structure (including the major cell types of the epithelium) of the seminiferous tubules.
SOM.MK.I.BPM2.2.ER.1.HCB.1085	Describe the major groups of testicular tumors.
SOM.MK.I.BPM2.2.ER.1.HCB.1086	Describe classical seminoma of the testis.
SOM.MK.I.BPM2.2.ER.1.HCB.1087	State the functions of the Sertoli cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1088	Discuss the microanatomical features of Sertoli cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1089	Describe the structure of the blood-testis barrier.
SOM.MK.I.BPM2.2.ER.1.HCB.1090	Identify Leydig cells by its microanatomical features and location.
SOM.MK.I.BPM2.2.ER.1.HCB.1091	State the function(s) of Leydig cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1092	Distinguish between the microscopic structure of spermatogonia, primary spermatocytes, secondary spermatocytes and spermatids.
SOM.MK.I.BPM2.2.ER.1.HCB.1093	Describe the phases of spermatogenesis (including spermatogonial, spermatocyte, spermatid phases).
SOM.MK.I.BPM2.2.ER.1.HCB.1094	Give the function(s) of spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1095	Describe the microscopic structure of spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1096	State the effect(s) of Kartagener’s syndrome on spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1097	State the anatomical location of the seminal vesicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1098	State the function(s) of the seminal vesicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1099	Describe the microscopic structure of the seminal vesicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1100	Identify the anatomical location of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1101	State the function(s) of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1102	Describe the microscopic structure (including corpora amylacea) of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1103	Discuss the secretions of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1104	Describe benign prostatic hyperplasia.
SOM.MK.I.BPM2.2.ER.1.HCB.1105	Describe adenocarcinoma of the prostate.
SOM.MK.I.BPM2.2.ER.1.HCB.1106	Give the anatomical location of the bulbourethral glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1107	State the function of the bulbourethral glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1108	Describe the microscopic structure of the bulbourethral glands.

DLA HCB	Histology of the Male Reproductive System
SOM.MK.I.BPM2.2.ER.1.HCB.1061	Describe the intratesticular tubular path travelled by spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1062	Describe the extratesticular tubular path travelled by spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1063	Give the function(s) of the tubuli recti.
SOM.MK.I.BPM2.2.ER.1.HCB.1064	Describe the microscopic structure of the tubuli recti.
SOM.MK.I.BPM2.2.ER.1.HCB.1065	Give the function(s) of the rete testis.
SOM.MK.I.BPM2.2.ER.1.HCB.1066	Describe the microscopic structure rete testis.
SOM.MK.I.BPM2.2.ER.1.HCB.1067	Give the function(s) of the efferent ductules.

SOM.MK.I.BPM2.2.ER.1.HCB.1068
SOM.MK.I.BPM2.2.ER.1.HCB.1069
SOM.MK.I.BPM2.2.ER.1.HCB.1070
SOM.MK.I.BPM2.2.ER.1.HCB.1071
SOM.MK.I.BPM2.2.ER.1.HCB.1072
SOM.MK.I.BPM2.2.ER.1.HCB.1073
SOM.MK.I.BPM2.2.ER.1.HCB.1074
SOM.MK.I.BPM2.2.ER.1.HCB.1075
SOM.MK.I.BPM2.2.ER.1.HCB.1076

Describe the microscopic structure of the efferent ductules.
Give the function(s) of the epididymis.
Describe the microscopic structure of the epididymis.
State the function of the ductus deferens.
Describe the histological appearance of the structure of the ductus deferens.
State the function of the erectile bodies of the penis.
Describe the histological appearance of the structure of the erectile bodies of the penis.
Describe the characteristics and composition of semen.
Distinguish histologically between the structure of the prostatic, membranous and penile (spongy) urethras.

Lecture PHYS
SOM.MK.II.BPM2.1.ER.1.PHYS.1065
SOM.MK.II.BPM2.1.ER.1.PHYS.1066

Male Reproductive Functions
Explain the role of the human male in reproduction and list the physiological functions of human male reproductive tract.
Describe the role of Sertoli cells, Leydig cells and the basement membrane in spermatogenesis
Explain the age-related changes in the male hypothalamo-pituitary-goandal axis that lead to puberty, reproductive maturity, and reproductive senescence (andropause).
Describe each of these hormones GnRH , FSH, LH and their mechanism of action
List the major target organs for testosterone and other androgens.
Explain the biosynthesis, mechanism of transport within the blood, metabolism and elimination of testosterone and related androgens.
Describe the neural, vascular, and endocrine components of the erection and ejaculation response.

Identify the causes and consequences of over secretion and under secretion of testosterone for a) prepubertal and b) postpubescent males c) infertility
Discuss the role of hypothalamic-gonadal axis in the regulation of hormone synthesis.

Lecture ANAT
SOM.MKII.BPM2.1.ER.1.ANAT.1020

SOM.MKII.BPM2.1.ER.1.ANAT.1119
SOM.MKII.BPM2.1.ER.1.ANAT.1131
SOM.MKII.BPM2.1.ER.1.ANAT.1022
SOM.MKII.BPM2.1.ER.1.ANAT.1023
SOM.MKII.BPM2.1.ER.1.ANAT.1024

SOM.MKII.BPM2.1.ER.1.ANAT.1026
SOM.MKII.BPM2.1.ER.1.ANAT.1027

Clinical Anatomy and Imaging of the Reproductive system - Female Reproductive Organs
Identify the major female pelvic organs, including the bladder, uterus, uterine tubes, ovaries, vaginal canal and rectum
Describe the anatomical relationships of the female pelvic organs to each other and surrounding structures, including the pelvic diaphragm and the bony pelvis
Identify the female pelvic organs on medical imaging studies such as MRI, CT scans, cystourethrography, hysterosalpingogram etc
Describe the ovaries and ovarian ligaments.
Describe the parts of the uterus and uterine tubes.
Define the terms anteverted, anteflexed, retroverted and retroflexed with respect to the position of the uterus.
Describe the broad ligament of the uterus and its subdivisions in relationship to the body and fundus of the uterus, the uterine tubes and their fimbriated ends, and the round ligament of the uterus.
Explain the clinical significance of the transverse cervical (cardinal), pubocervical and uterosacral ligaments in uterine support.

SOM.MKII.BPM2.1.ER.1.ANAT.1029	Identify the uterus and uterine tubes on a hyeterosalpingogram and it’s clinical significance
SOM.MKII.BPM2.1.ER.1.ANAT.1021	Describe the reflections of the peritoneum over the bladder, uterus, uterine tubes, ovaries, vaginal canal and rectum in the pelvic cavity.
SOM.MKII.BPM2.1.ER.1.ANAT.1025	Describe the typical relationship of the cervix to the vaginal canal.
SOM.MKII.BPM2.1.ER.1.ANAT.1031	Describe the parts and relationships of the vagina.
SOM.MKII.BPM2.1.ER.1.ANAT.1037	Explain the clinical importance of the relationship of the posterior fornix to the rectouterine pouch.
SOM.MKII.BPM2.1.ER.1.ANAT.1033	Describe the blood supply, venous and lymphatic drainage of the ovaries, uterine tubes, fundus, body and cervix of uterus, pelvic part of vagina, vaginal vestibule.
SOM.MKII.BPM2.1.ER.1.ANAT.1034	Describe the course of the uterine arteries across the pelvic floor.
SOM.MKII.BPM2.1.ER.1.ANAT.1035	Describe “the water under the bridge” relationship and use this information to discuss surgical / anatomical considerations during hysterectomy.
SOM.MKII.BPM2.1.ER.1.ANAT.1127	Explain how fluid in the abdomen can reach the female pelvic peritoneal pouches and where it would ultimately settle.
SOM.MKII.BPM2.1.ER.1.ANAT.1121	Identify the female pelvic organs on medical imaging studies such as MRI, CT scans, etc.
SOM.MKII.BPM2.1.ER.1.ANAT.1028	Define the term “uterine prolapse”, list the structures that are compromised and explain why it is more common in multiparous women who delivered via the vaginal canal.
SOM.MKII.BPM2.1.ER.1.ANAT.1032	Discuss the importance of vaginal and rectal examinations and “pap smears”.
SOM.MKII.BPM2.1.ER.1.ANAT.1030	Explain how lower genitourinary infections in females, such as gonorrhea and chlamydia, can gain access to the pelvic peritoneal spaces.
DLA ANAT	Autonomic Innervation of the Female Pelvic Organs
SOM.MKII.BPM2.1.ER.1.ANAT.1049	Describe the origin, and type of nerve fibers, found in the superior hypogastric plexus and the inferior hypogastric (pelvic) plexus
SOM.MKII.BPM2.1.ER.1.ANAT.1120	Describe the autonomic innervation of the female pelvic organs and the relationship to the pelvic pain line
SOM.MKII.BPM2.1.ER.1.ANAT.1039	Explain the differences in sensory innervation of the pelvic part of the vagina versus the perineal part of the vagina.
SOM.MKII.BPM2.1.ER.1.ANAT.1128	Compare and contrast the visceral pain pathways for structures above the peritoneal floor of the pelvis versus structures below the peritoneal floor (pelvic pain line) in females
Lecture ANAT	Clinical Anatomy and Imaging of the Reproductive system - Female Perineum
SOM.MKII.BPM2.1.ER.1.ANAT.1122	Describe the contents of the female urogenital triangle
SOM.MKII.BPM2.1.ER.1.ANAT.1123	Describe the superficial perineal fascia in females (Colles’ fascia) and its relationship with the superficial fascia of the abdomen and labia, urethral opening and vaginal orifice
SOM.MKII.BPM2.1.ER.1.ANAT.1124	Describe the deep perineal fascia in females
SOM.MKII.BPM2.1.ER.1.ANAT.1052	List the boundaries, contents and innervation of the deep perineal pouch in female
SOM.MKII.BPM2.1.ER.1.ANAT.1053	List the boundaries, contents and innervation of the superficial perineal pouch in female
SOM.MKII.BPM2.1.ER.1.ANAT.1051	Describe the structure and innervation of the female external genitalia (Vulva) including the erectile bodies, clitoris, labia majora, labia minora, vestibule of the vagina

SOM.MKII.BPM2.1.ER.1.ANAT.1125	Describe the anatomical location, structure and function of Bartholin’s (greater vestibular) glands. Describe how blockage of the Bartholin gland ducts can lead to cyst formation and how infection of the cyst can lead to the development of a Bartholin gland abscess.
SOM.MKII.BPM2.1.ER.1.ANAT.1126	
SOM.MKII.BPM2.1.ER.1.ANAT.1013	Describe the arterial supply, venous drainage and lymphatic drainage of the female perineum.
SOM.MKII.BPM2.1.ER.1.ANAT.1014	Describe the course of the pudendal nerve and its branches within the female perineum.
SOM.MKII.BPM2.1.ER.1.ANAT.1015	Describe the course of the internal pudendal artery in the female pelvis and perineum.
SOM.MKII.BPM2.1.ER.1.ANAT.1010	Describe an oblique episiotomy and it’s clinical indication.
SOM.MKII.BPM2.1.ER.1.ANAT.1007	List the boundaries of the anal triangle.
SOM.MKII.BPM2.1.ER.1.ANAT.1008	Describe the ischioanal fossa, its location, contents, innervation and clinical significance.
SOM.MKII.BPM2.1.ER.1.ANAT.1116	Describe a perianal abcess and its relationship to the ischioanal fossa
SOM.MKII.BPM2.1.ER.1.ANAT.1129	Describe the innervation of the female perineum
SOM.MKII.BPM2.1.ER.1.ANAT.1133	Identify the parts of the female perineum on imaging studies suchs as CT scans and MRI

Lecture HCB

SOM.MK.I.BPM2.2.ER.1.HCB.1113	Histology of the Female Reproductive System - Ovary to Oviduct Give the anatomical location of the ovaries.
SOM.MK.I.BPM2.2.ER.1.HCB.1114	Describe the blood supply of the ovaries.
SOM.MK.I.BPM2.2.ER.1.HCB.1115	Briefly describe the structural support of the ovaries.
SOM.MK.I.BPM2.2.ER.1.HCB.1116	Give the function(s) of the ovaries.
SOM.MK.I.BPM2.2.ER.1.HCB.1117	Describe the microscopic structure of the ovaries.
SOM.MK.I.BPM2.2.ER.1.HCB.1118	Discuss the hormonal regulation of the ovary.
SOM.MK.I.BPM2.2.ER.1.HCB.1119	Discuss the menstrual cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1120	Discuss the ovarian cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1121	Describe the histological changes on the endometrium during the menstrual cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1122	Describe the histological changes on the breasts during the menstrual cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1123	Describe the histological changes on the endometrium during the ovarian cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1124	Describe the histological changes on the breasts during the ovarian cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1125	List the phases of development of the ovarian follicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1126	State the role of follicle stimulating hormone (FSH) on the ovarian follicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1127	Describe the process of development of the ovum within the follicles. Distinguish between primordial, primary, secondary and tertiary (mature/Graafian) follicles based on their microscopic structure, function and location within the ovary.
SOM.MK.I.BPM2.2.ER.1.HCB.1128	
SOM.MK.I.BPM2.2.ER.1.HCB.1129	State the function(s) of the zona pellucida.
SOM.MK.I.BPM2.2.ER.1.HCB.1130	Describe the microscopic structure of the zona pellucida.
SOM.MK.I.BPM2.2.ER.1.HCB.1131	Identify the granulosa cells of the follicle.

SOM.MK.I.BPM2.2.ER.1.HCB.1132
SOM.MK.I.BPM2.2.ER.1.HCB.1133
SOM.MK.I.BPM2.2.ER.1.HCB.1134
SOM.MK.I.BPM2.2.ER.1.HCB.1135
SOM.MK.I.BPM2.2.ER.1.HCB.1136
SOM.MK.I.BPM2.2.ER.1.HCB.1137
SOM.MK.I.BPM2.2.ER.1.HCB.1138
SOM.MK.I.BPM2.2.ER.1.HCB.1139
SOM.MK.I.BPM2.2.ER.1.HCB.1140
SOM.MK.I.BPM2.2.ER.1.HCB.1141
SOM.MK.I.BPM2.2.ER.1.HCB.1142
SOM.MK.I.BPM2.2.ER.1.HCB.1143
SOM.MK.I.BPM2.2.ER.1.HCB.1144
SOM.MK.I.BPM2.2.ER.1.HCB.1145
SOM.MK.I.BPM2.2.ER.1.HCB.1146
SOM.MK.I.BPM2.2.ER.1.HCB.1147
SOM.MK.I.BPM2.2.ER.1.HCB.1148
SOM.MK.I.BPM2.2.ER.1.HCB.1149
SOM.MK.I.BPM2.2.ER.1.HCB.1150
SOM.MK.I.BPM2.2.ER.1.HCB.1151
SOM.MK.I.BPM2.2.ER.1.HCB.1152
SOM.MK.I.BPM2.2.ER.1.HCB.1153
SOM.MK.I.BPM2.2.ER.1.HCB.1154
SOM.MK.I.BPM2.2.ER.1.HCB.1155
SOM.MK.I.BPM2.2.ER.1.HCB.1156
SOM.MK.I.BPM2.2.ER.1.HCB.1157

Give the function(s) of the granulosa cells.
Describe the microscopic structure of the granulosa cells.
State the role of follicle stimulating hormone (FSH) on the granulosa cells.
Discuss the development of the theca folliculi: theca interna and theca externa.
Discuss the function(s) of the cells of the theca interna.
Describe the microscopic structure of the cells of the theca interna.
State the role of luteinizing hormone (LH) on the theca interna cells.
Discuss the relationship between the theca intena cells and the granulosa cells
State the location of atretic follicles.
Describe atretic follicles.
State the role of luteinizing hormone (LH) on the ovarian follicles.
Describe the sequence of events that lead to ovulation.
Describe the microscopic structure of the corpus luteum.
State the origin of the granulosa lutein cells of the corpus luteum.
Identify the hormones produced by the granulosa lutein cells of the corpus luteum.
State the origin of the theca lutein cells of the corpus luteum.
Identify the hormones produced by the theca lutein cells of the corpus luteum.
State the function(s) of the of corpus luteum during pregnancy.
State the function(s) of the of corpus luteum during the menstrual cycle.
Describe the stages of the development of corpus luteum during the menstrual cycle.
Describe the stages of the development of the corpus luteum during pregnancy.
State the function(s) of the oviduct (uterine tube).
Describe the microscopic structure of the oviduct (uterine tube).
State the function(s) of the uterus.
Describe the microscopic structure (including the layers) of the uterine wall.
Trace the blood supply of the uterus.

ER SG 04 - Male Reproductive system

Small group HCB;PHYS;ANAT
SOM.MK.II.BPM2.1.ER.1.PHYS.1404
SOM.MK.II.BPM2.1.ER.1.PHYS.1405
SOM.MK.II.BPM2.1.ER.1.PHYS.1406
SOM.MK.II.BPM2.1.ER.1.PHYS.1407
SOM.MK.I.BPM2.2.ER.1.HCB.1109
SOM.MK.I.BPM2.2.ER.1.HCB.1110

Create a flow chart to explain the role of testosterone in various stages of life in males, starting from intrauterine stage.
Differentiate the neural, vascular, and endocrine components of the erection and ejaculation response.
Summarize the role of hypothalamo-pituitary-testicular axis in the regulation of testicular functions.
Differentiate primary and secondary hypo and hypergonadism with respect to their clinical presentations and laboratory findings.
Identify and describe the cells, tissues, organs and associated structures of the Male reproductive system.
Explain the key roles of the cells and organs associated with the Male reproductive system to the group.

SOM.MK.I.BPM2.2.ER.1.HCB.1111	Apply the histology of the Male reproductive system to clinical cases and correlations.
SOM.P.I.BPM2.2.ER.6.HCB.1112	Develop both professional and presentation skills.
SOM.MK.II.BPM1.1.ER.1.ANAT.5001	Analyze the blood supply, venous drainage, and lymphatic drainage of the prostate gland, ductus deferens, seminal vesicles, bulbourethral glands, testes, scrotum, and penis, assessing their roles in maintaining the function of the male reproductive system.
SOM.MK.II.BPM1.1.ER.1.ANAT.5002	Trace the pathway of the vas deferens from the epididymis, through the inguinal canal, into the pelvic cavity, and to its termination at the ejaculatory ducts, evaluating its anatomical significance in male reproductive function.
SOM.MK.II.BPM1.1.ER.1.ANAT.5003	Examine the autonomic innervation of the male pelvic organs, analyzing how these nerves relate to the pelvic pain line and affect organ function.
SOM.MK.II.BPM1.1.ER.1.ANAT.5004	Describe the relationship of the prostate gland to the ampulla of the ductus deferens, seminal vesicles, bladder, urethra, rectum, and pelvic floor, discussing the anatomical context and clinical implications of these connections.
SOM.MK.II.BPM1.1.ER.1.ANAT.5005	Evaluate the anatomical basis for the symptoms commonly associated with benign prostatic enlargement and prostate cancer, identifying the key structures involved in symptom manifestation.
SOM.MK.II.BPM1.1.ER.1.ANAT.5006	Analyze how prostate cancer leads to erectile and urinary dysfunction, explaining the role of affected nerves and discussing potential clinical outcomes.
SOM.MK.II.BPM2.1.ER.1.ANAT.5001	Analyze the blood supply, venous drainage, and lymphatic drainage of the prostate gland, ductus deferens, seminal vesicles, bulbourethral glands, testes, scrotum, and penis, assessing their roles in maintaining the function of the male reproductive system.
SOM.MK.II.BPM2.1.ER.1.ANAT.5002	Trace the pathway of the vas deferens from the epididymis, through the inguinal canal, into the pelvic cavity, and to its termination at the ejaculatory ducts, evaluating its anatomical significance in male reproductive function.
SOM.MK.II.BPM2.1.ER.1.ANAT.5003	Examine the autonomic innervation of the male pelvic organs, analyzing how these nerves relate to the pelvic pain line and affect organ function.
SOM.MK.II.BPM2.1.ER.1.ANAT.5004	Describe the relationship of the prostate gland to the ampulla of the ductus deferens, seminal vesicles, bladder, urethra, rectum, and pelvic floor, discussing the anatomical context and clinical implications of these connections.
SOM.MK.II.BPM2.1.ER.1.ANAT.5005	Evaluate the anatomical basis for the symptoms commonly associated with benign prostatic enlargement and prostate cancer, identifying the key structures involved in symptom manifestation.
SOM.MK.II.BPM2.1.ER.1.ANAT.5006	Analyze how prostate cancer leads to erectile and urinary dysfunction, explaining the role of affected nerves and discussing potential clinical outcomes.

Lecture HCB

SOM.MK.I.BPM2.2.ER.1.HCB.1158	Histology of the Uterus, Vagina & Breast Development
SOM.MK.I.BPM2.2.ER.1.HCB.1159	State the function of the cervix.
SOM.MK.I.BPM2.2.ER.1.HCB.1160	Describe the histological appearance of the structure of the cervix.
SOM.MK.I.BPM2.2.ER.1.HCB.1161	Describe the histological changes on the cervix due to the hormones produced during the ovarian cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1162	Describe the histological changes on the cervix due to the hormones produced during the menstrual cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1163	State the function of the vagina.
	Describe the histological appearance of the structure of the vagina.

SOM.MK.I.BPM2.2.ER.1.HCB.1164	Describe the histological changes on the vagina due to the hormones produced during the ovarian cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1165	Describe the histological changes on the vagina due to the hormones produced during the menstrual cycle.
SOM.MK.I.BPM2.2.ER.1.HCB.1166	State the function(s) of the placenta.
SOM.MK.I.BPM2.2.ER.1.HCB.1167	Discuss some of the hormones produced by the placenta.
SOM.MK.I.BPM2.2.ER.1.HCB.1168	Briefly discuss the development of the placenta.
SOM.MK.I.BPM2.2.ER.1.HCB.1169	Differentiate between the microscopic features of the primary, secondary and tertiary chorionic villi of the placenta.
SOM.MK.I.BPM2.2.ER.1.HCB.1170	State the function(s) of the mammary glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1171	Describe the microscopic structure of the mammary glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1172	Distinguish between the histological structure of inactive, active, lactating and menopausal mammary glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1173	Describe the effects of hormones on the mammary glands during development.
SOM.MK.I.BPM2.2.ER.1.HCB.1174	Describe the effects of hormones on the mammary glands during menstruation.
SOM.MK.I.BPM2.2.ER.1.HCB.1175	Describe effects of hormones on the mammary glands during pregnancy.
SOM.MK.I.BPM2.2.ER.1.HCB.1176	Describe effects of hormones on the mammary glands during lactation.
SOM.MK.I.BPM2.2.ER.1.HCB.1177	Describe effects of hormones on the mammary glands during menopause.
SOM.MK.I.BPM2.2.ER.1.HCB.1178	Describe the changes seen in the uterus due to fibroids.
SOM.MK.I.BPM2.2.ER.1.HCB.1179	Discuss the significance of transformation zone of the cervix.
SOM.MK.I.BPM2.2.ER.1.HCB.1180	Describe the changes in the ovary as seen in polycystic ovarian disease.
SOM.MK.I.BPM2.2.ER.1.HCB.1181	Explain the clinical significance of glycogen content within vaginal epithelial cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1182	Briefly describe the changes seen in mammary glands in adenocarcinoma.

Lecture PHYS
SOM.MK.II.BPM2.1.ER.1.PHYS.1074

SOM.MK.II.BPM2.1.ER.1.PHYS.1075	Explain the roles of GnRH, FSH, LH, estradiol, and inhibin in oogenesis and follicular maturation in females at puberty, the start of ovarian cycle
SOM.MK.II.BPM2.1.ER.1.PHYS.1076	Describe the hormonal regulation of estrogen and progesterone biosynthesis and secretion by the ovary.
SOM.MK.II.BPM2.1.ER.1.PHYS.1077	Describe the the hormonal changes, follicular development, ovulation, formation & atresia of corpus luteum throughout the ovarian cycle
SOM.MK.II.BPM2.1.ER.1.PHYS.1078	Describe the physiological actions of estrogen on its target organs.
SOM.MK.II.BPM2.1.ER.1.PHYS.1079	List the principal physiological actions of progesterone, its major effects on target organs.
SOM.MK.II.BPM2.1.ER.1.PHYS.1080	Describe how the changes in ovarian steroids produce the proliferative and secretory phases of the uterine endometrium and menstruation and the changes in basal body temperature during the menstrual cycle
SOM.MK.II.BPM2.1.ER.1.PHYS.1081	Explain the age-related changes in the hypothalamo-pituitary-goandal axis in females that lead to puberty, reproductive maturity, and reproductive senescence (menopause).
SOM.MK.II.BPM2.1.ER.1.PHYS.1082	Explain the physiological basis of steroid hormone contraception.

SOM.MK.II.BPM2.1.ER.1.PHYS.1083	Explain clinical features of hypogonadism during different ages in females.
Lecture PHYS	Physiology of Pregnancy
SOM.MK.II.BPM2.1.ER.1.PHYS.1084	Describe the role of human chorionic gonadotropin (hCG) in the rescue of the corpus luteum in maintaining pregnancy early post-implantation
SOM.MK.II.BPM2.1.ER.1.PHYS.1085	List the hormones secreted by the placenta and discuss the roles of sex steroids, and other hormones during of pregnancy
SOM.MK.II.BPM2.1.ER.1.PHYS.1086	Describe the interactions between the placenta and the fetus in the pathway for production of estrogens during pregnancy.
SOM.MK.II.BPM2.1.ER.1.PHYS.1087	Explain the maternal and fetal endocrinology and physiological changes during pregnancy.
SOM.MK.II.BPM2.1.ER.1.PHYS.1088	Explain preeclampsia and eclampsia associated with pregnancy.
SOM.MK.II.BPM2.1.ER.1.PHYS.1089	Explain how gestational diabetes can lead to problems during pregnancy.
SOM.MK.II.BPM2.1.ER.1.PHYS.1093	Explain the basis for the inhibition of milk secretion during pregnancy and the initiation of lactation after parturition.
DLA PHYS	Parturition & Lactation
SOM.MK.II.BPM2.1.ER.1.PHYS.1090	Discuss the interaction of fetal and mechanical factors in initiation of parturition.
SOM.MK.II.BPM2.1.ER.1.PHYS.1142	Discuss the role of hormones in initiation and progression of parturition.
SOM.MK.II.BPM2.1.ER.1.PHYS.1091	Discuss some causes of delayed or prolonged labor
SOM.MK.II.BPM2.1.ER.1.PHYS.1092	Explain the role of hormones in mammary gland development during puberty, pregnancy, and lactation.
SOM.MK.II.BPM2.1.ER.1.PHYS.1094	Describe the neuroendocrine regulation of milk secretion and milk ejection & role of prolactin in contributing to anovulatory cycle
Lecture ANAT	Development of the Reproductive system
SOM.MKII.BPM2.1.ER.1.ANAT.1101	Describe the time-line for development of the genital system.
SOM.MKII.BPM2.1.ER.1.ANAT.1079	Describe the development of the gonads and the differentiation into testes and ovaries.
SOM.MKII.BPM2.1.ER.1.ANAT.1093	Discuss the level at which the gonads develop and its relationship to referred pain.
SOM.MKII.BPM2.1.ER.1.ANAT.1080	Describe the descent of the testes into the scrotum and the clinical significance of failure to descend.
SOM.MKII.BPM2.1.ER.1.ANAT.1083	Describe the development of the male reproductive ducts and associated glands.
SOM.MKII.BPM2.1.ER.1.ANAT.1084	Describe the development of the female reproductive ducts in relation to the paramesonephric ducts, uterovaginal primordium and urogenital sinus.
SOM.MKII.BPM2.1.ER.1.ANAT.1087	Describe the formation and differentiation of the external genitalia of males and females from the urogenital folds and labioscrotal swelling.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1464	Describe the development, location, blood supply, innervation and venous drainage of the suprarenal glands.
DLA ANAT	Fetal Membranes
SOM.MKII.BPM2.1.ER.1.ANAT.1068	Describe the formation of the placenta.
SOM.MKII.BPM2.1.ER.1.ANAT.1069	Describe the fetomaternal junction.

SOM.MKII.BPM2.1.ER.1.ANAT.1070	Describe the placental circulation.
SOM.MKII.BPM2.1.ER.1.ANAT.1071	Describe how the amniotic fluid is produced and its circulation.
SOM.MKII.BPM2.1.ER.1.ANAT.1072	Describe the development of allantois and fetal membranes.
SOM.MKII.BPM2.1.ER.1.ANAT.1073	Describe placenta accreta, percreta and previa and their clinical significance
SOM.MKII.BPM2.1.ER.1.ANAT.1074	Describe hydatiform mole and its association with choriocarcinoma
SOM.MKII.BPM2.1.ER.1.ANAT.1075	Compare and contrast oligohydramnios versus polyhydramnios and describe their associated conditions
SOM.MKII.BPM2.1.ER.1.ANAT.1076	Describe the cause of hemolytic disease in a newborn (erythroblastosis fetalis)
SOM.MKII.BPM2.1.ER.1.ANAT.1077	Describe embryogenesis of monozygotic, dizygotic twins and conjoined twins
SOM.MKII.BPM2.1.ER.1.ANAT.1078	Describe the mechanism of twin to twin transfusion

Lecture GNET	Genetic Screening
	Define genetic screening, identify its objectives, different types of genetic screening, differentiate between population based vs. targeted screening, and discuss the difference between screening tests Vs diagnostic tests
SOM.MK.III.BPM2.4.ER.3.GNET.1012	Discuss the applications of cfDNA testing from maternal serum and the importance of next-generation sequencing in the test.
SOM.MK.III.BPM2.4.ER.3.GNET.1013	Differentiate between population screening programs, particularly Prenatal and Neonatal screening programs, what are the markers used and the importance of timing
SOM.MK.III.BPM2.4.ER.3.GNET.1014	Evaluate some existing targeted screening programs for groups of people at high risk for certain genetic diseases: PKU; Thalassemia; Sickle cell disease; Cystic fibrosis
SOM.MK.III.BPM2.4.ER.3.GNET.1015	Summarize the requirements of a genetic screening program
SOM.MK.III.BPM2.4.ER.3.GNET.1016	Define “genetic counselling” and identify the founding principles
SOM.MK.III.BPM2.4.ER.3.GNET.1017	Define the role of a genetic counselor in the face of a positive diagnosis: accuracy of the tests, calculation of reoccurrence risk
SOM.MK.III.BPM2.4.ER.3.GNET.1018	
SOM.MK.III.BPM2.4.ER.3.GNET.1019	Appreciate the need to provide appropriate assistance in the decision-making process of patients when faced with a positive result and further testing

DLA GNET	Developmental Genetics HOX Genes & Achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1005	Describe the HOX gene
SOM.MK.III.BPM2.4.ER.3.GNET.1006	Illustrate the function of the HOX genes
SOM.MK.III.BPM2.4.ER.3.GNET.1007	Discuss disorders associated with HOX genes
SOM.MK.III.BPM2.4.ER.3.GNET.1008	Describe the general features of achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1009	Discuss the inheritance pattern of the achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1010	Explain the possible outcomes to the offspring of two parents who both have achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1011	Analyze the effect of the FGFR3 G380R mutation and explain why the disorder can be classified as autosomal dominant to gain of function

DLA GNET
SOM.MK.III.BPM2.4.ER.3.GNET.1001
SOM.MK.III.BPM2.4.ER.3.GNET.1002
SOM.MK.III.BPM2.4.ER.3.GNET.1003
SOM.MK.III.BPM2.4.ER.3.GNET.1004

Lecture ANAT
SOM.MKII.BPM2.1.ER.1.ANAT.1081
SOM.MKII.BPM2.1.ER.1.ANAT.1082
SOM.MKII.BPM2.1.ER.1.ANAT.1088
SOM.MKII.BPM2.1.ER.1.ANAT.1089
SOM.MKII.BPM2.1.ER.1.ANAT.1090
SOM.MKII.BPM2.1.ER.1.ANAT.1091

SOM.MKII.BPM2.1.ER.1.ANAT.1085
SOM.MKII.BPM2.1.ER.1.ANAT.1086
SOM.MKII.BPM2.1.ER.1.ANAT.1092

DLA ANAT

SOM.MKII.BPM2.1.ER.1.ANAT.1099
SOM.MKII.BPM2.1.ER.1.ANAT.1100

DLA ANAT

SOM.MKII.BPM2.1.ER.1.ANAT.1094
SOM.MKII.BPM2.1.ER.1.ANAT.1095

SOM.MKII.BPM2.1.ER.1.ANAT.1096
SOM.MKII.BPM2.1.ER.1.ANAT.1097

SOM.MKII.BPM2.1.ER.1.ANAT.1098

Small group HCB;GNET;ANAT;PHYS; BCHI

Developmental Genetics Terminology & Sonic Hedgehog

Discuss the vocabulary associated with congenital dysmorphology and give examples.
Describe the SHH signalling pathway
Discuss how SHH functions to establish cell fate during development
Describe disorders associated with defects of SHH signalling

Clinical Embryology of the Reproductive System

Describe the anatomical location, mechanism and clinical consequences in the development of hydrocele testis.
Describe defects in testicular descent including cryptorchidism and ectopic testes.
Explain the developmental mechanism responsible for the formation of hypospadias and epispadias.
Describe the identification and clinical significance of disorders of sex development (DSD’s)
Describe the consequence of decreased levels of dihydrotestosterone on development of male characteristics.
Describe the cause and clinical picture of androgen insensitivity syndrome.
Describe the development of birth defects of the female genital tract including a double uterus, a unicornuate uterus, a bicornuate uterus vaginal agenesis and imperforate hymen.
Describe the anatomical location, mechanism and clinical sequences of canal of Nuck hernias.
Describe the formation and clinical significance of Gartner duct cysts.

Fetal Monitoring

Describe the procedures for assessing fetal status including diagnostic amniocentesis, chorionic villus sampling, percutaneous umbilical cord blood sampling, alpha fetoprotein assay and fetal monitoring.
Describe the significance of alpha fetoprotein assay.

Teratogenesis

Describe the developmental anomalies induced after infection by rubella virus, varicella virus, cytomegalovirus, toxoplasma gondii, treponema pallidum, herpes simplex
Describe the congenital defects associated with maternal diabetes.
Describe the mechanism and clinical presentation for congenital anomalies associated with in utero exposure to the following: cigarette smoke, alcohol ingestion, tetracyclines, retinoic acid, warfarin, phenytoin, valproic acid, cocaine, organic mercury
Describe the developmental anomalies associated with thalidomide.
List and identify the locations where expression of fibroblast growth factor is diminished after exposure to thalidomide and those associated with phocomelia.

ER SG 05 - Female Reproductive system

SOM.MK.III.BPM2.1.ER.2.BCHM.1417	Evaluate the significance of nutrition in pregnancy, lactation and the infancy.
SOM.MK.III.BPM2.1.ER.2.BCHM.1418	Explain the effects of nutritional factors during pregnancy that affect fetal development
SOM.MK.II.BPM2.1.ER.1.PHYS.1408	Prepare a graphic represetation of changes in the ovarian and anterior pituitary hormones during different stages of a menstrual cycle and correlate them with the corresponding alterations in the ovaries and uterine endometrium.
SOM.MK.II.BPM2.1.ER.1.PHYS.1409	Summarize the role of hypothalamo-pituitary-ovarian axis in the regulation of ovarian functions during puberty, reproductive period and post-menupausal period in females.
SOM.MK.II.BPM2.1.ER.1.PHYS.1410	Design a graphic representation of various hormones secreted during different stages of gestation and relate them to their sources and functions.
SOM.MK.II.BPM2.1.ER.1.PHYS.1411	Outline the physiological changes in the mother during pregnancy and correlate them with the maternal and fetal welbeing.
SOM.MK.III.BPM2.1.ER.2.GNET.0401	Compare and contrast the different types of non-invasive and invasive prenatal screening tests and the identifiy the importance of optimal timing for the tests so that the markers which are used are informative of the fetal status
SOM.MK.III.BPM2.1.ER.2.GNET.0402	Distinguish between first and second trimester maternal serum screenings tests and findings which predict Trisomy 21, Trisomy 18, Trisomy 13 and neural tube defects and describe which disorders also have significant ultrasound findings
SOM.MK.I.BPM2.2.ER.1.HCB.1183	Identify and describe the cells, tissues, organs and associated structures of the Female reproductive system.
SOM.MK.I.BPM2.2.ER.1.HCB.1184	Present and demonstrate the cells, tissues, organs and associated structures of the Female reproductive system to the group.
SOM.MK.I.BPM2.2.ER.1.HCB.1185	Apply the histology of the Female reproductive system to clinical cases and correlations.
SOM.P.I.BPM2.5.ER.6.HCB.1186	Mastering of professional skills and presentation skills.
SOM.MK.II.BPM1.1.ER.1.ANAT.5007	Evaluate the procedures for assessing fetal status, analyzing diagnostic methods such as amniocentesis, chorionic villus sampling, percutaneous umbilical cord blood sampling, alpha fetoprotein assay, and fetal monitoring, and discussing their clinical significance and indications.
SOM.MK.II.BPM1.1.ER.1.ANAT.5008	Analyze the arterial supply, venous drainage, and lymphatic drainage of the female perineum, assessing their functional and clinical relevance in maintaining perineal health.
SOM.MK.II.BPM1.1.ER.1.ANAT.5009	Examine the innervation of the female perineum, evaluating the role of nerves in sensory and motor functions within the region.
SOM.MK.II.BPM1.1.ER.1.ANAT.5010	Explain the procedure of an oblique episiotomy and evaluate its anatomic implications.
Lecture BCHM	Nutritional Aspects of Pregnancy, Lactation and Infant Nutrition
SOM.MK.III.BPM2.1.ER.2.BCHM.1062	List the consequences of maternal malnutrition on: Maternal health, Fetal health and the infant during lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1063	Review the need for nutritional balance and the consequences of deficiencies on mother, fetus and infant as it relates to: Energy requirements and weight gain (Including carbohydrates and Omega 3 fatty acids like DHA), Proteins (Including enzymes and immunoglobulins in breast milk), Iron (Microcytic iron deficiency anemia), Folate (Macrocytic anemia and neural tube defects), Calcium, Vitamin A, Zinc and Iodine (congenital hypothyroidism).
SOM.MK.III.BPM2.1.ER.2.BCHM.1064	Explain the importance of safe handling and preparation of food, especially during pregnancy and lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1065	Discuss the effects and list some compounds that may pose health risks during pregnancy and lactation like Caffeine, Smoking, Alcohol (fetal alcohol syndrome), Drugs: medical and non-medical, Toxins and teratogens.

SOM.MK.III.BPM2.1.ER.2.BCHM.1066	Discuss the effect of alcohol on the fetus and newborn including fetal alcohol syndrome.
SOM.MK.III.BPM2.1.ER.2.BCHM.1067	Review the significance of screening for gestational diabetes in the expectant mother and explain the complications of maternal hyperglycemia on fetal development including fetal macrosomia.
SOM.MK.III.BPM2.1.ER.2.BCHM.1068	Evaluate the need to monitor weight of the expectant mother, infant and the lactating mother and provide counseling to achieve pre-pregnancy weight.
SOM.MK.III.BPM2.1.ER.2.BCHM.1069	List specific macronutrients required for neonate and infant growth.
SOM.MK.III.BPM2.1.ER.2.BCHM.1070	Discuss lactose content in human breast milk. Explain significance of milk exclusion in congenital lactase deficiency and in classical galactosemia.
SOM.MK.III.BPM2.1.ER.2.BCHM.1071	Describe the function of human milk oligosaccharides (HMOs) regarding the GI tract and immune system in infants. Discuss the change of serum levels of immunoglobulins in the fetus and infant.
SOM.MK.III.BPM2.1.ER.2.BCHM.1072	Discuss the importance of DHA and arachidonic acid in breast milk related to brain growth and eye development
SOM.MK.III.BPM2.1.ER.2.BCHM.1073	Compare and contrast composition of cow's milk and breast milk.
SOM.MK.III.BPM2.1.ER.2.BCHM.1074	Highlight the significance of dietary essential amino acids and the conditionally essential amino acids needed in infants (positive nitrogen balance).
DLA BCHM	Nutritional Aspects of Lactation and Infant Nutrition
SOM.MK.III.BPM2.1.ER.2.BCHM.1055	Highlight the major differences in the composition of colostrum, transition milk and mature milk during lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1056	Discuss the advantages of human milk compared to animal milk and infant formula.
SOM.MK.III.BPM2.1.ER.2.BCHM.1057	Review WHO guidelines about breastfeeding and HIV.
SOM.MK.III.BPM2.1.ER.2.BCHM.1058	Identify the age range of neonate and infant. Discuss Vitamin K deficiency bleeding (VKDB) in the neonate and infant.
SOM.MK.III.BPM2.1.ER.2.BCHM.1059	Identify micronutrients that are low in human breast milk. Highlight the dietary modifications (iron) to be implemented after 6 months of age.
SOM.MK.III.BPM2.1.ER.2.BCHM.1060	Describe bioavailability of iron and factors that reduce inorganic iron uptake.
Lecture BCHM	Flipped Class: Nutritional Disorders of Adolescence & Childhood
SOM.MK.III.BPM2.1.ER.2.BCHM.1076	Define and differentiate the following conditions: Nutrition, wasting, cachexia, marasmus, kwashiorkor, stunting.
SOM.MK.III.BPM2.1.ER.2.BCHM.1077	Describe malnutrition in children and its various forms.
SOM.MK.III.BPM2.1.ER.2.BCHM.1078	Describe anorexia nervosa.
SOM.MK.III.BPM2.1.ER.2.BCHM.1079	Describe childhood overnutrition including risk factors, methods of assessment, basis of management and complications
SOM.MK.III.BPM2.1.ER.2.BCHM.1080	Define undernutrition with respect to macronutrients and micronutrients.
SOM.MK.III.BPM2.1.ER.2.BCHM.1081	Compare and contrast Kwashiorkor, Marasmus and Marasmic-kwashiorkor based on clinical features, lab findings and anthropometric measurements.
SOM.MK.III.BPM2.1.ER.2.BCHM.1082	Explain associated micronutrient deficiency with special reference to calcium, phosphate, magnesium, iodine and selenium.

SOM.MK.III.BPM2.1.ER.2.BCHM.1083 Highlight the complications that may occur in the management of children with undernutrition and the basis for nutritional supplementation.